CertTalk: A Clear, Proof-First Protocol for Two Agents With Private Maps

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Abstract
Two agents each see only part of a grid's obstacles. They must decide if a route exists from start s to goal t in the union map U—and hand over a proof anyone can replay later. CertTalk does not ship maps; it ships a certificate. If a path exists, the certificate is the path. If no path exists, the certificate is a small set of blocked cells that separates s from t. This writeup presents the idea in plain language, shows two worked examples, and prints full, annotated transcripts for the baselines so a reader can verify what each system actually sends.

1 What We Decide (and What We Don't)

Problem. A knows obstacles A. B knows obstacles B. Neither sees the union $U = A \cup B$. The question is: does a 4-neighbour path exist from s to t in U?

Contract. The dialogue ends with a single certificate that a third-party checker (the "oracle") can replay using only the certificate, s, t, grid size, and U. The oracle never reads the negotiation transcript and never needs the agents' planner code.

Out of scope. No motion control, dynamics, or multi-agent coordination beyond the yes/no reachability proof. Certificates are short (target ≤ 256 bytes) and archivable.

2 How the Proof Works

Path certificate (checker's steps).

1. Decode Base64 runs (RLE4 bytes).
2. Verify `digest16` (CRC16 of raw bytes).
3. Expand runs into coordinates from s to t; check 4-connectivity and bounds.
4. Confirm no coordinate lies in a blocked cell of U.
5. Accept. The bytes *are* the proof.

Cut certificate (checker's steps).

1. Decode Base64 cell list (DELTA16) and witness bits.
2. Verify `digest16` (CRC16 of raw bytes).
3. Check each listed cell is blocked in U.
4. Remove the cells from the underlying 4-neighbour graph; verify s and t are disconnected.
5. Accept. The bytes *are* the proof.

Witness bits. One bit per cut cell says "who saw this cell" (0=A, 1=B). They guide the responder during negotiation. The oracle ignores witness semantics except to check length.

3 Worked Example: Reachable (seed 247)

When a path exists, A proposes a concrete route. B replays it locally. If every step is free for B, B upgrades it to a certificate. Three messages end the dialogue.

CertTalk — PATH success


```
{
  "t": "PATH_PROPOSE",
  "s": "v1", "q": 0,
  "p": { "runs": "QoVBgkKCRA==", "digest16": 22270 },
  "c": 16269
}
{
  "t": "PATH_CERT",
  "s": "v1", "q": 0,
  "p": { "runs": "QoVBgkKCRA==", "digest16": 22270, "signed_by": ["B"] },
  "c": 49459
}
{
  "t": "ACK",
  "s": "v1", "q": 1,
  "p": { "ack_of": "PATH_CERT", "digest16": 22270 },
  "c": 3754
}
```


What B checks. Decode runs; verify CRC; expand steps; confirm every coordinate is free in B's map; certify. The certificate itself is the replayable proof.

4 Worked Example: Unreachable (seed 145)

When no path exists, A proposes a short list of blocked cells that separates s from t. B validates per-cell (using witness attribution to allow "I saw it" from A) and then validates disconnection. Three messages suffice on this seed.

CertTalk — CUT success

```
{
  "t": "CUT_PROPOSE",
  "s": "v1", "q": 0,
  "p": { "cs": "AAACAAH/Af8=", "k": 3, "digest16": 11880, "w": "AA==" },
  "c": 51694
}
{
  "t": "CUT_CERT",
  "s": "v1", "q": 0,
  "p": { "cs": "AAACAAH/Af8=", "k": 3, "digest16": 11880, "w": "AA==", "sb": ["B"] },
  "c": 43260
}
{
  "t": "ACK",
  "s": "v1", "q": 1,
  "p": { "ack_of": "CUT_CERT", "digest16": 11880 },
  "c": 27687
}
```

What B checks. Decode cells and witness; verify CRC; accept each cell (blocked locally or credited to A); remove the set and confirm s and t are disconnected; certify.

5 Baselines: What They Send (and Why It Matters)

Different systems reach the same decision with different bytes. Below we show one clean unreachable seed (145) where all four succeed, and why a reader should care.

Send-All — CUT success (seed 145)

```
{
  "type": "PROBE", "schema": "v1", "seq": 0,
  "payload": { "what": "CELLS",
    "indices":
        [2,3,4,11,13,14,17,19,20,22,27,32,36,37,38,39,46,47,63,65,68,70,71,78,80,83,85,95,98],

    "chunk": 0, "total": 1, "final": true },
  "crc16": 40037
}
{
  "type": "CUT_CERT", "schema": "v1", "seq": 0,
  "payload": { "cells": "AgAAAP4BAQA=", "encoding": "DELTA16_v1",
    "digest16": 34702, "k": 3, "witness": "Ag==", "signed_by": ["B"] },
  "crc16": 12256
}
{
  "type": "ACK", "schema": "v1", "seq": 1,
  "payload": { "ack_of": "CUT_CERT", "digest16": 34702 },
  "crc16": 57117
}
```

Why it matters. Send-All may stream raw blocked indices first. The later certificate—not the raw indices—is the replayable proof.

Greedy-Probe — CUT success (seed 145) Same as CertTalk above on this clean seed (three messages: CUT_PROPOSE, CUT_CERT, ACK). Probes appear only when needed.

Responder-MinCut — CUT success (seed 145)


```
{
  "t": "SCHEMA", "s": "v1", "q": 0,
  "p": { "cut_enc": "DELTA16_v1", "path_enc": "RLE4_v1", "s": [0,0], "size": [10,10], "t": [9,9] },
  "c": 64959
}
{
  "t": "CUT_CERT", "s": "v1", "q": 0,
  "p": { "cs": "BwAHAAEAAQD9Af8B", "k": 5, "digest16": 4126, "w": "Hw==", "sb": ["B"] },
  "c": 32263
}
{
  "t": "NACK", "s": "v1", "q": 1,
  "p": { "n": "CUT_PROPOSE", "reason": "BLOCKED", "x": [[7,7],[9,7],[6,8]] },
  "c": 593
}
{
  "t": "PROBE", "s": "v1", "q": 1,
  "p": { "cs": [[7,7],[9,7],[6,8]], "wht": "CELLS" },
  "c": 55063
}
{
  "t": "HELLO", "s": "v1", "q": 2, "p": {}, "c": 26486
}
{
  "t": "DONE", "s": "v1", "q": 2, "p": {}, "c": 22322
}
```


Why it matters. Shows the longer path to agreement when the first cut is insufficient; one small probe reconciles the disagreement.

6 Results in One Table

Medians across 1,000 cached instances (schema-free run).

System	Bytes	Rounds
Send-All	364	3
CertTalk	468	5
Greedy-Probe	468	5
Responder-MinCut	506	6

Takeaway: CertTalk spends about one extra hundred bytes versus Send-All, but those bytes *are the proof* (path or cut), which simplifies auditing.

7 Limits and What to Expect

- Two agents; 10×10 grids; reliable in-order channel. Larger grids may require chunked certificates.
- No privacy model: witness bits reveal who attested each cut cell. Signatures are optional if authenticity is required.
- The checker never reads the transcript; it only replays the final artifact with U.